

SHYAM SUNDER

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Professional Summary

AI/ML Engineer with hands-on experience building **machine learning, deep learning, NLP, GenAI, and MLOps systems** from data ingestion and feature engineering to model training, evaluation, experiment tracking, API deployment, and cloud infrastructure. Strong production experience with **Python, PyTorch, TensorFlow, FastAPI, MLflow, Docker, AWS, LangChain, RAG, SQL, and MongoDB**, delivering measurable improvements in accuracy, automation, reporting, and operational efficiency.

Skills

- **Languages:** Python, SQL, Go, C
- **AI/ML:** Machine Learning, Deep Learning, NLP, Computer Vision, PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, Embeddings
- **GenAI/LLMs:** LLM Applications, RAG, Prompt Engineering, LangChain, LangGraph, Google Gemini, OpenAI API
- **MLOps/Cloud:** MLflow, ONNX, Docker, CI/CD, AWS (ECR/ECS/S3/Lambda), Databricks, Ansible
- **Backend/Data:** FastAPI, Flask, REST APIs, JWT, MongoDB, SQL, Data Pipelines, Model Serving

Experience

McDermott International

July 2024-Present

Automation Engineer

Chennai

Autoreport – Internal Data Analytics System

Python, Pandas, Flask, Excel Automation

- Built an automated **data analysis and reporting system** generating structured Word and Excel reports, reducing report preparation time by **20%** and saving more than **10,000 man-hours**.
- Designed **data ingestion, logging, and archival pipelines** for ML-ready datasets.
- Deployed the system on an internal server with **multi-user LAN access**.
- Developed a lightweight **Flask-based frontend** for non-technical users.

Pipe Support Recommendation System (Digital Twin – In Development)

PyTorch, PINNs, MLflow, FastAPI, Docker

- Built **ML-ready data pipelines** to extract, clean, validate, and standardize legacy SACS (Bentley) engineering data for AI model development.
- Extracted **5,000+ distinct pipe supports** across **10 different parameters** to build a standardized training dataset.
- Developing a **digital twin recommendation engine** using machine learning and physics-informed modeling to reduce engineering workload and rental costs.
- Designing **Physics-Informed Neural Networks (PINNs)** with custom physics-aware loss functions for constraint-driven model training.
- Implemented **ANN baselines in PyTorch**, integrated physics constraints, and compared architectures for improved model generalization.
- Set up **MLflow** for experiment tracking, metric comparison, model registry, and reproducible ML lifecycle management.
- Developing a **containerized FastAPI inference service** with Docker for scalable model serving and API-based predictions.
- Built an **SQL database and Flask dashboard** to upload raw SACInput files, automatically extract and preprocess pipe support data, and enable retrieval, filtering, and statistical analysis.

Projects

Athena: Sentiment Analysis – End-to-End MLOps System

NLP, PyTorch, MLflow, FastAPI, Docker, AWS, Databricks, ONNX

- Built a **production-grade MLOps pipeline** for NLP sentiment analysis covering data ingestion, preprocessing, model training, evaluation, MLflow tracking, Docker containerization, and AWS deployment, reducing manual release effort by $\sim 70\%$.
- Improved model performance by integrating **pretrained text embeddings**, achieving $\sim 20\%$ **relative accuracy improvement** over baseline models while mitigating overfitting using regularization.
- Implemented **Andrej Karpathy's AutoResearch framework** to run multi-agent experiments across different model architectures in Athena, delivering an additional $\sim 10\%$ **relative accuracy gain**.
- Implemented **embedding caching and feature reuse** to avoid redundant recomputation, reduce training time, and improve experimentation velocity.
- Tracked and versioned models using **MLflow**, managing **10+ experiments** and multiple model versions.
- Migrated MLflow tracking from self-hosted EC2/S3 to **Databricks** to improve scalability and reliability.
- Optimized inference by exporting models to **ONNX**, reducing Docker image size by $\sim 20\%$ and lowering cold-start latency.
- Built a **FastAPI model inference service** with a lightweight frontend for real-time sentiment prediction and API-based model serving.
- Automated **CI/CD-driven container builds**, resolved **ARM vs AMD64** architecture issues, and deployed on Amazon ECS via Amazon ECR with **zero-downtime updates**.

Argus: EfficientNet Image Recognition System

Computer Vision, PyTorch, CNN, EfficientNet, ONNX, Flask, Docker

- Implemented **EfficientNet from scratch in PyTorch** by reproducing the research paper and validating computer vision model correctness.
- Resolved **pretrained weight incompatibility** by designing a custom parameter-mapping strategy.
- Implemented and evaluated **CNN, MobileNet, and EfficientNet** architectures for image classification benchmarking and comparative analysis.
- Exported trained models to **ONNX** for framework-agnostic inference.
- Built a **Flask-based REST API** and Dockerized the service for **self-hosted deployment** on an Ubuntu server.

Enterprise RAG System: Secure Document Intelligence Platform

RAG, LangGraph, LangChain, Google Gemini, FAISS, Streamlit, RBAC, Python

- Built a **production-grade Retrieval-Augmented Generation (RAG) system** with **role-based access control (RBAC)** enforcing document-level permissions across **4 user roles**: admin, analyst, auditor, and viewer.
- Designed a **7-stage LangGraph pipeline** covering injection detection, RBAC filtering, query routing, FAISS retrieval, context merging, LLM generation, and citation formatting.
- Implemented **multi-format document ingestion** for PDF, CSV, JSON/JSONL, and plain text with incremental indexing using **MD5 change detection**.
- Integrated **Google Gemini embeddings and generation** with a custom LangChain-compatible embeddings wrapper using concurrent API calls via **ThreadPoolExecutor**.
- Built a **prompt injection defense layer** with **13+ regex-based detection patterns**, query sanitization, control character filtering, and dual RBAC enforcement before and after retrieval.
- Developed an interactive **CLI** using Click and prompt_toolkit plus a **Streamlit web UI** with chat interface, admin dashboard, source citations, provenance tracing, and relevance scores.
- Secured authentication with **PBKDF2-SHA256 password hashing**, constant-time comparison, file-level permission hardening, and data-leakage prevention across role boundaries.

Achhilife.in: Psychological Assessment SaaS

GenAI, FastAPI, MongoDB, LangChain, Google Gemini, AWS Lambda, Ansible

- Built a **RESTful FastAPI backend** for a psychological assessment SaaS with **MongoDB**, supporting multi-tenant workflows for guests, merchants, practitioners, and admins across **30+ endpoints**.
- Integrated **Google Gemini 2.5 Flash Lite** via **LangChain** to analyze psychometric responses and generate AI-powered clinical insights, including severity, score, summary, and recommendations.
- Implemented secure authentication with **JWT, httpOnly cookies, bcrypt passwords, OTP-verified call requests, and role-based access control** following OWASP-aligned practices.

- Designed a **multi-tenant merchant affiliate system** with slug-based branding, fork-on-edit question architecture, and isolated patient data per practitioner portal.
- Deployed the application to **AWS Lambda** using the **Mangum ASGI adapter** and to a **Hetzner VPS**, automating server provisioning with **Ansible playbooks**.

Achhicv: GenAI Based CV Enhancement

GenAI, RAG, FastAPI, LangChain, LangGraph, MongoDB, JWT

- Built a **FastAPI-based backend** for a CV management and ATS optimization system.
- Integrated **LLM-based workflows** using LangChain and OpenAI APIs.
- Orchestrated a **LangGraph workflow** to extract all CV information in a single end-to-end pipeline.
- Designed **secure authentication** using JWT access and refresh tokens, implementing login and signup flows.
- Integrated **MongoDB** for structured CV storage, user data persistence, and retrieval workflows.
- Integrated **RAG pipelines** for document extraction, semantic retrieval, and AI-generated CV creation.
- Integrated Amazon S3 for storing and retrieval of documents.

Education

St. Stephen’s College, University of Delhi

Sep. 2019 – May 2022

BSc (H) Physics

University of Delhi, Delhi

Indian Institute of Technology, Madras

Jul. 2022 – May 2024

MSc. Physics

IITM, Chennai